An Internship Report on

Machine Learning and Edge Computing enabled Energy Management System for optimization of Hybrid Electric Powertrain: Proof-of-Concept and Predictive EMS

Submitted partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in

Artificial Intelligence and Machine Learning

By

Mishael Julian (2462184)

Under the Guidance of

Dr. Sujatha A K

Department of AI and Data Science Engineering
School of Engineering and Technology,
CHRIST (Deemed to be University),
Kumbalagudu, Bengaluru - 560 074

May 2026

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CHRIST (Deemed to be University) is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment.

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Faith in God
Moral Uprightness
Love of Fellow Beings
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"To excel in human-centred AI and data-driven innovation through ethical research, societal well-being, and transformative collaborations.To excel in human-centred AI and data-driven innovation through ethical research, societal well-being, and transformative collaborations."

Department Mission

M1: *Empowering individuals to ethically harness data and AI through accessible and value-driven curriculum.*

M2: *Department Foster a dynamic research environment that advances innovative and impactful solutions for the betterment of global well-being.*

M3: *Innovate scientific knowledge and Entrepreneurship through academia and Industry collaborations.*

Program Educational Objectives (PEOs)

PEO1: *Professional Acumen: Understand, analyze and design solutions with professional competency for the real-world problems.*

PEO2: *Critical Analysis: Develop software/embedded solutions for the requirements, based on critical analysis and research.*

PEO3: *Team work: Function effectively in a team and as an individual in a multidisciplinary/multicultural environment*

PEO4: *Life Long Learning: Accomplish holistic development comprehending professional responsibilities.*

Program Outcomes (POs)

PO1: Engineering Knowledge *Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop the solution of complex engineering problems.*

PO2: Problem Analysis *Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.*

PO3: Design/Development of Solutions *Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.*

PO4: Conduct Investigations of Complex Problems *Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.*

PO5: Engineering Tool Usage *Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.*

PO6: The Engineer and The World *Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.*

PO7: Ethics *Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.*

PO8: Individual and Collaborative Team work *Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.*

PO9: Communication *Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.*

PO10: Project Management and Finance *Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.*

PO11: Life-Long Learning *Recognize the need for, and have the preparation and ability for (i) independent and life-long learning (ii) adaptability to new and emerging technologies and (iii) critical thinking in the broadest context of technological change.*

Program Specific Outcomes (PSOs)

PSO1: *PSO1.*

PSO2: *PSO2*

PSO3: *PSO3*

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School of Engineering and Technology

Department of AI and Data Science Engineering

CERTIFICATE

This is to certify that **Mishael Julian (2462184)** has successfully completed his/her internship work entitled "**Machine Learning and Edge Computing enabled Energy Management System for optimization of Hybrid Electric Powertrain: Proof-of-Concept and Predictive EMS**" at **CHRIST University, Mysore Rd, Kanmanike, Kengeri, Kumbalgodu, Bengaluru, Karnataka 560074** , from **4th May** to **th June** for a duration of **32** days and submitted on «Internship Viva Date» in partial fulfillment for the award of **Bachelor of Technology** in **Artificial Intelligence and Machine Learning** during the academic year **2026–27**.

GUIDE
Dr. Sujatha A K
(Signature with Date)

HEAD OF THE DEPARTMENT
Dr. Michael Moses T
(Signature with Seal)

EXAMINER 1
(Name & Signature with Date)

Examiner 2
(Name & Signature with Date)

ACKNOWLEDGEMENT

We would like to thank Dr Rev Fr Joseph C. C., Vice Chancellor, CHRIST (Deemed to be University), Dr Rev Fr Viju P. D., Pro Vice Chancellor, Dr Fr Jiby Jose, Director, School of Engineering and Technology, Fr Shijin P. J., Assistant Director, School of Engineering and Technology, CHRIST (Deemed to be University), Dr Raghunandan Kumar R., Dean, and Dr Mary Anitha E. A., Associate Dean, for their kind patronage.

We would also like to express sincere gratitude and appreciation to **«HOD Name»**, Head of the Department, **Department of AI and Data Science Engineering** for giving me this opportunity to take up this project.

We also extremely grateful to my guide, **Dr. Sujatha A K**, who has supported and helped to carry out the project. His constant monitoring and encouragement helped me keep up to the project schedule.

We also extremely grateful to my co-guide, **«Industry Guide Name»**, who has supported and helped to carry out the project. His constant monitoring and encouragement helped me keep up to the project schedule.

If outside the college-mention the organisation and the concerned people, like head of the organisation, guide and any other person you want to thank. All faculty and non- teaching staff. You may acknowledge your parents or any who supported you.

DECLARATION

We hereby declare that the project titled **"Machine Learning and Edge Computing enabled Energy Management System for optimization of Hybrid Electric Powertrain: Proof-of-Concept and Predictive EMS"** is a record of the original internship work undertaken as part of the course **«SubjectCode – SubjectName»**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology**.

We completed this internship under the supervision of **Dr. Sujatha A K** and **«Industry Guide Name»**, at **«Company Name», «Company Address»**, during the period from **«Internship Start Date»** to **«Internship End Date»**.

We further declare that this internship report has not been submitted, either in whole or in part, for the award of any degree, diploma, associateship, fellowship, or any other academic qualification at any other institution. Furthermore, this report has not been submitted for publication or presentation elsewhere.

We affirm that this internship work has been carried out in accordance with the Code of Research Conduct prescribed by the University.

Place: CHRIST (Deemed to be University), Kumbalagudu, Bengaluru - 560 074
Date: «Declaration Date»
Signature: Mishael Julian (2462184)

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CHAPTER 1

INTRODUCTION

The automotive industry is undergoing a rapid transformation toward sustainable and intelligent transportation systems. Hybrid Electric Vehicles (HEVs) have emerged as an effective solution for reducing fuel consumption and greenhouse gas emissions while maintaining the driving range and reliability of conventional internal combustion engine vehicles. A key component of every HEV is the Energy Management System (EMS), which determines how power is distributed between the internal combustion engine, electric motor, and battery under varying driving conditions.

Traditional rule-based Energy Management Systems rely on predefined control strategies that are unable to anticipate future traffic conditions, route characteristics, or driver demand. Recent advances in Artificial Intelligence (AI) and Machine Learning (ML) provide opportunities to develop predictive control systems capable of improving fuel economy, battery utilization, regenerative braking performance, and overall vehicle efficiency.

This internship was carried out as part of Project PHEX '27 (Predictive Hybrid Energy Executive), an academic research initiative focused on developing an AI-assisted supervisory Energy Management System for Hybrid Electric Vehicles. The work involved designing and implementing predictive control modules, traffic-aware decision-making, demand forecasting algorithms, State-of-Charge (SOC) planning strategies, rule-based supervisory control, and simulation-based performance evaluation using Python. The developed Proof-of-Concept demonstrates how predictive intelligence can be integrated with conventional EMS strategies to improve overall hybrid vehicle operation while maintaining system safety and reliability.

1.1 Objectives

The primary objectives of the internship were:

- Study the architecture and operation of Hybrid Electric Vehicle Energy Management Systems.
- Design a supervisory rule-based Energy Management System for intelligent power management.
- Develop predictive demand forecasting models using Machine Learning techniques.
- Implement traffic-aware route prediction for proactive energy management.
- Design a State-of-Charge (SOC) planning strategy for battery optimization.
- Develop an intelligent decision engine capable of selecting optimal operating modes.

- Integrate safety override mechanisms to ensure reliable EMS operation.
- Validate the proposed EMS through simulation and performance evaluation using multiple Key Performance Indicators (KPIs).

1.2 Company / Project Profile

Project PHEX '27 (Predictive Hybrid Energy Executive) is a student-led research and development initiative undertaken under the guidance of faculty mentors from CHRIST (Deemed to be University). The project focuses on developing an intelligent Energy Management System for Hybrid Electric Vehicles by integrating Artificial Intelligence, Machine Learning, predictive control algorithms, and vehicle simulation technologies.

The objective of Project PHEX '27 is to build a scalable Proof-of-Concept capable of predicting future driving conditions and optimizing energy distribution between the engine, electric motor, and battery. The project combines traditional rule-based supervisory control with predictive decision-making models to improve fuel economy, battery utilization, regenerative braking efficiency, and overall vehicle performance. During this internship, the primary contribution involved the development of predictive EMS modules, including demand forecasting, route-aware SOC planning, intelligent mode selection, safety override logic, simulation workflow, performance benchmarking, and technical documentation.

CHAPTER 2

TECHNICAL DESCRIPTION AND IMPLEMENTATION

2.1 Actual Work

During the internship, the primary responsibility was the development of an AI-assisted supervisory Energy Management System (EMS) for Hybrid Electric Vehicles under Project PHEX '27. The implementation combined conventional rule-based control with predictive machine learning techniques to improve battery utilization, fuel economy, and overall vehicle efficiency. The work involved designing predictive control modules, implementing simulation models, integrating subsystem communication, and validating the complete EMS through multiple drive-cycle simulations.

2.1.1 Prerequisites

- Strong programming knowledge in Python.
- Understanding of Hybrid Electric Vehicle (HEV) architecture.
- Knowledge of Energy Management Systems (EMS).
- Familiarity with Machine Learning concepts and Scikit-learn.
- Understanding of NumPy, Pandas, and Matplotlib.
- Knowledge of predictive modelling and Random Forest regression.
- Basic understanding of vehicle dynamics and battery management.
- Familiarity with Git-based software development and modular Python programming.

2.1.2 Responsibilities

- Task completion within deadlines
- Daily reporting to mentor
- Independent problem-solving

2.1.3 Challenges

Dealing with missing values, outliers, inconsistencies, and feature selection were major challenges. Imputation strategies required careful consideration.

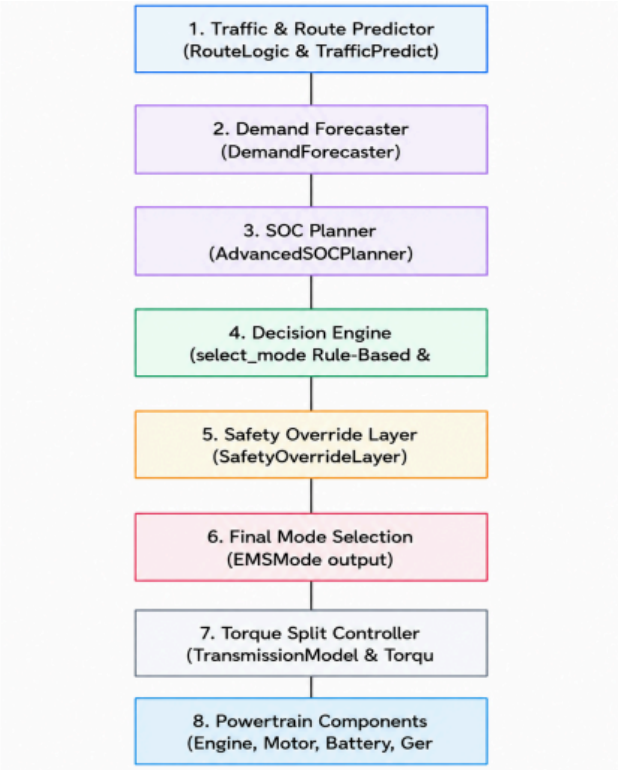


Figure 2.1: Project PHEX '27 Supervisory Energy Management System Architecture

2.1.4 EMS Performance Summary

Table 2.1: EMS Performance Summary

Performance Metric	Value	Unit
Fuel Economy	1.27	L/100 km
SOC Deviation (RMS)	1.86	%
Energy Recuperation	92.0	%
EV Mode Share	39.9	%
Hybrid Mode Share	13.8	%
ICE Mode Share	17.3	%
Operating Cost	0.0358	Currency/km
CO_2 Emissions	29.3	g/km
Average ICE Efficiency	33.2	%
Peak Battery Power	22.17	kW

2.1.5 Vehicle Power Demand

The Demand Forecaster estimates the power required to move the vehicle under changing road and driving conditions. The total traction force is expressed as

$$F_{\text{traction}} = F_{\text{rolling}} + F_{\text{aero}} + F_{\text{grade}} + F_{\text{acceleration}}, \tag{2.1}$$

and the corresponding vehicle power demand is

$$P_{\text{demand}} = F_{\text{traction}}\, v, \tag{2.2}$$

where P_{demand} is the required vehicle power in watts, F_{traction} is the total traction force in newtons, and v is the vehicle speed in metres per second. The traction force comprises rolling resistance, aerodynamic drag, road-grade resistance, and acceleration force.

2.1.6 Rule-Based Supervisory EMS Algorithm

The implemented `select_mode()` function applies rule-based supervisory control to select EV, Hybrid, or ICE operation while accounting for vehicle demand, battery state, traffic conditions, route context, and safety constraints.

Algorithm 1 Rule-Based Supervisory EMS Mode Selection Algorithm

```python
def select_mode(
    telemetry: dict,
    traffic_state: TrafficState,
    soc_plan: SOCPlan,
    battery_health: BatteryHealthState,
    predicted_demand_kw: float,
    regen_result: RegenResult,
) -> EMSMode:
    speed = telemetry.get("speed_kmh", 0.0)
    soc = telemetry.get("soc", 0.0)
    soc_frac = soc / 100.0 if soc > 1.0 else soc
    power_demand = telemetry.get("power_demand_kw", 0.0)
    throttle = telemetry.get("throttle", 0.0)
    current_segment = telemetry.get("current_segment", "urban")
    next_segment = telemetry.get("next_segment", "urban")
    position = telemetry.get("position_km", 0.0)
    rpm = telemetry.get("rpm", 0.0)
    # 1. Regenerative Braking
    if power_demand < 0.0:
        if soc_frac >= MAX_SOC:
            logger.info("RULE6_REGEN: Regen disabled - batte
        else:
            clamped_regen = min(abs(power_demand), MOTOR_PEA
            # 1 Wh = 1/3600 kWh
            recovered_wh = clamped_regen * (1.0 / 3600.0) * 1000.0
            logger.info("RULE6_REGEN: Regenerative braking active. Recovered Wh: %.4f", recovered_wh)
            return EMSMode.REGEN
    # 2. Maximum Acceleration
    if throttle >= 0.90:
        logger.info("RULE7_MAX_ACCEL: Maximum acceleration requested.")
        return EMSMode.HYBRID_ASSIST
    # 3. Low SOC Recovery
    if soc_frac <= 0.25:
        logger.info("RULE8_SOC_RECOVERY: Low SOC (SOC=%.2f) recovery active.", soc_frac)
        return EMSMode.CHARGE_SUSTAIN
    # 4. Urban Stop-and-Go EV Priority
    if speed < 50.0 and (traffic_state == TrafficState.CONGESTED or traffic_state.value == "CONGESTED") and current_segment.lower() == "urban":
        if soc_frac <= (MIN_SOC + 0.05):
            logger.info("RULE1_SUPPRESSED_LOW_SOC: Urban EV Priority suppressed due to low SOC (SOC=%.2f)", soc_frac)
        else:
            logger.info("RULE1_EV_PRIORITY: Urban stop-and-go prioritising EV operation.")
            return EMSMode.EV
    # 5. SOC Preservation Before Urban Zones
    upcoming_urban = next_segment.lower() in ("urban", "stop_go")
    if upcoming_urban and soc_frac < 0.80 and current_segment.lower() in ("highway", "mixed", "arterial"):
        logger.info("RULE2_SOC_PRESERVE: Upcoming urban zone - preserving SOC (SOC=%.2f).", soc_frac)
        return EMSMode.CHARGE_SUSTAIN
    # 6. Traffic-Aware Mode Planning
    if traffic_state == TrafficState.CONGESTED or traffic_state.value == "CONGESTED":
        logger.info("RULE3_TRAFFIC_AWARE: High congestion forecast - biasing EV/Hybrid mode.")
        return EMSMode.HYBRID_ASSIST
    # 7. GPS Route Preview Mode Planning
    if current_segment.lower() == "urban":
        logger.info("RULE4_ROUTE_PREVIEW: Urban segment - favouring EV mode.")
        return EMSMode.EV
    elif current_segment.lower() == "highway":
        logger.info("RULE4_ROUTE_PREVIEW: Highway segment - favouring ICE efficiency mode.")
        return EMSMode.ICE
    # 8. ICE Efficiency Sweet Spot (C.RULE5)
    if rpm > 0.0 and (rpm < 2000.0 or rpm > 3000.0):
        logger.info("RULE5_RPM_SWEET_SPOT: Engine outside peak band (%d RPM). Adjusting split.", int(rpm))
        if rpm > 3000.0:
            # Shed load
            return EMSMode.HYBRID_ASSIST
        else:
            # Deactivate or increase load
            return EMSMode.EV
```

(attribute) __mul__: float

Return self*value.

Go to float

CHAPTER 3

CONCLUSION AND FUTURE SCOPE

3.1 Conclusion

The internship provided practical experience in the design and implementation of an intelligent Supervisory Energy Management System (EMS) for a Hybrid Electric Vehicle (HEV). The work involved developing predictive demand forecasting, route-aware State-of-Charge (SOC) planning, rule-based decision logic, safety override mechanisms, and torque-split control using Python. These modules were integrated into a unified simulation framework capable of selecting optimal operating modes under different driving conditions.

Simulation results demonstrated that the proposed EMS effectively balanced battery utilization, engine efficiency, regenerative braking, and overall energy consumption across multiple drive-cycle scenarios. The project also strengthened practical knowledge in machine learning, control systems, software architecture, data preprocessing, model evaluation, and modular software development. Overall, the internship provided valuable exposure to real-world automotive software development and intelligent vehicle energy management.

3.2 Future Scope

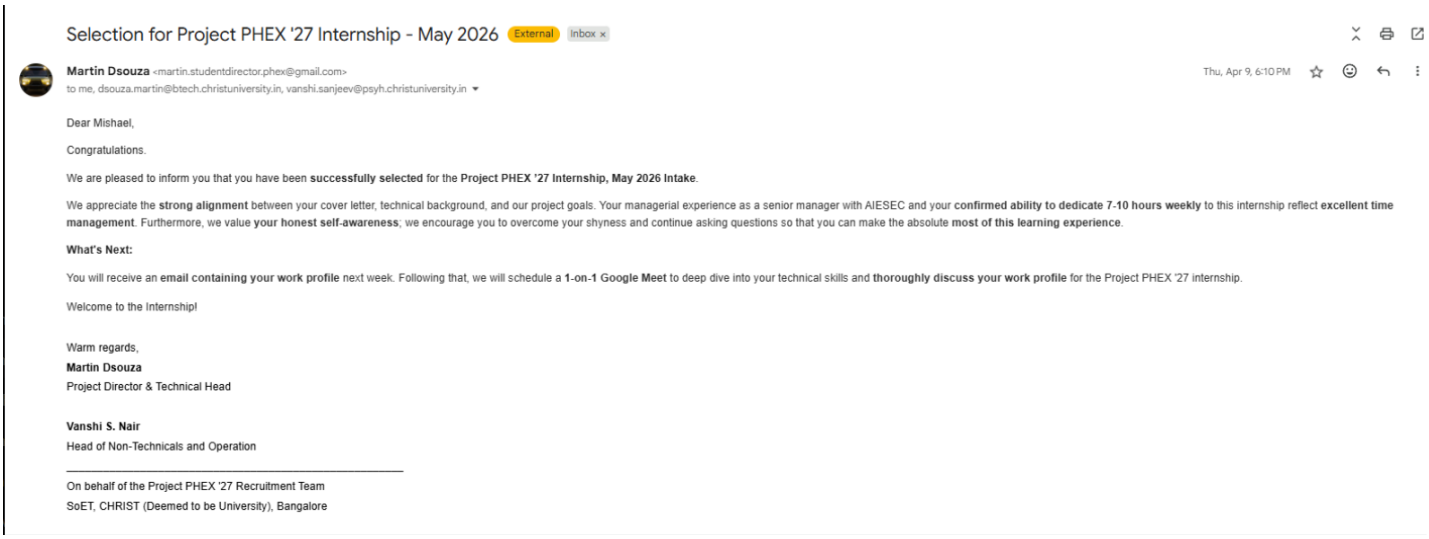
The developed Proof-of-Concept can be extended in several directions to support future phases of Project PHEX:

- Integration with real-time GPS navigation and live traffic information.
- Deployment of the EMS on embedded automotive hardware (ECU/SoC) for real-time execution.
- Implementation of CAN/CAN FD communication for interaction with vehicle subsystems.
- Replacement of rule-based decision logic with Reinforcement Learning or Model Predictive Control (MPC).
- Validation using industry-standard drive cycles such as WLTP, NEDC, FTP-75, and real-world telemetry.
- Integration of battery degradation prediction and thermal management models.
- Hardware-in-the-Loop (HIL) and Software-in-the-Loop (SIL) testing before vehicle deployment.

- Development of a graphical driver interface for monitoring energy flow, battery SOC, and operating mode.
- Extension of the framework to Plug-in Hybrid Electric Vehicles (PHEVs) and Battery Electric Vehicles (BEVs).
- Optimization for production-grade embedded systems with low-latency inference and real-time safety monitoring.

APPENDICES

A.1 Offer Letter



Selection for Project PHEX '27 Internship - May 2026 External Inbox ×

Martin Dsouza <martin.studentdirector.phex@gmail.com> Thu, Apr 9, 6:10 PM
to me, dsouza.martin@btech.christuniversity.in, vanshi.sanjeev@psyh.christuniversity.in

Dear Mishael,

Congratulations.

We are pleased to inform you that you have been **successfully selected** for the **Project PHEX '27 Internship, May 2026 Intake**.

We appreciate the **strong alignment** between your cover letter, technical background, and our project goals. Your managerial experience as a senior manager with AIESEC and your **confirmed ability to dedicate 7-10 hours weekly** to this internship reflect **excellent time management**. Furthermore, we value **your honest self-awareness**; we encourage you to overcome your shyness and continue asking questions so that you can make the absolute **most of this learning experience**.

What's Next:

You will receive an **email containing your work profile** next week. Following that, we will schedule a **1-on-1 Google Meet** to deep dive into your technical skills and **thoroughly discuss your work profile** for the Project PHEX '27 internship.

Welcome to the Internship!

Warm regards,
Martin Dsouza
Project Director & Technical Head

Vanshi S. Nair
Head of Non-Technicals and Operation

On behalf of the Project PHEX '27 Recruitment Team
SoET, CHRIST (Deemed to be University), Bangalore

A.2 Completion Certificate

CHRIST (Deemed to be University)

School of Engineering and Technology | Kengeri, Bengaluru, Karnataka, India

PROJECT PHEX '27

Machine Learning & Edge Computing enabled Energy Management System for Optimization of Hybrid Electric Powertrain

CERTIFICATE OF COMPLETION

This is to certify that

MISHAEL JULIAN

Reg. No. 2462184

has successfully completed the internship under Project PHEX '27, contributing to the design and validation of a Proof-of-Concept Hybrid Electric Vehicle Energy Management System, encompassing predictive EMS concepts, machine-learning-assisted energy optimization frameworks, drive-cycle simulation, and technical documentation, during the period 04 May 2026 to 06 June 2026, in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science Engineering with Specialization in Artificial Intelligence and Machine Learning.

Mr. Martin Dsouza
Student Project Technical Head
Project PHEX '27

Dr. Parag Jose
Faculty Co-ordinator &
Faculty Head COE E-mobility

Dr. Pradeep Kumar G.S.
Faculty Co-ordinator
Co-ordinator Automotive Technology

Date Submitted: _____

A.3 Code and Screenshots

A.3.1 Loading and Preparing Telemetry Data

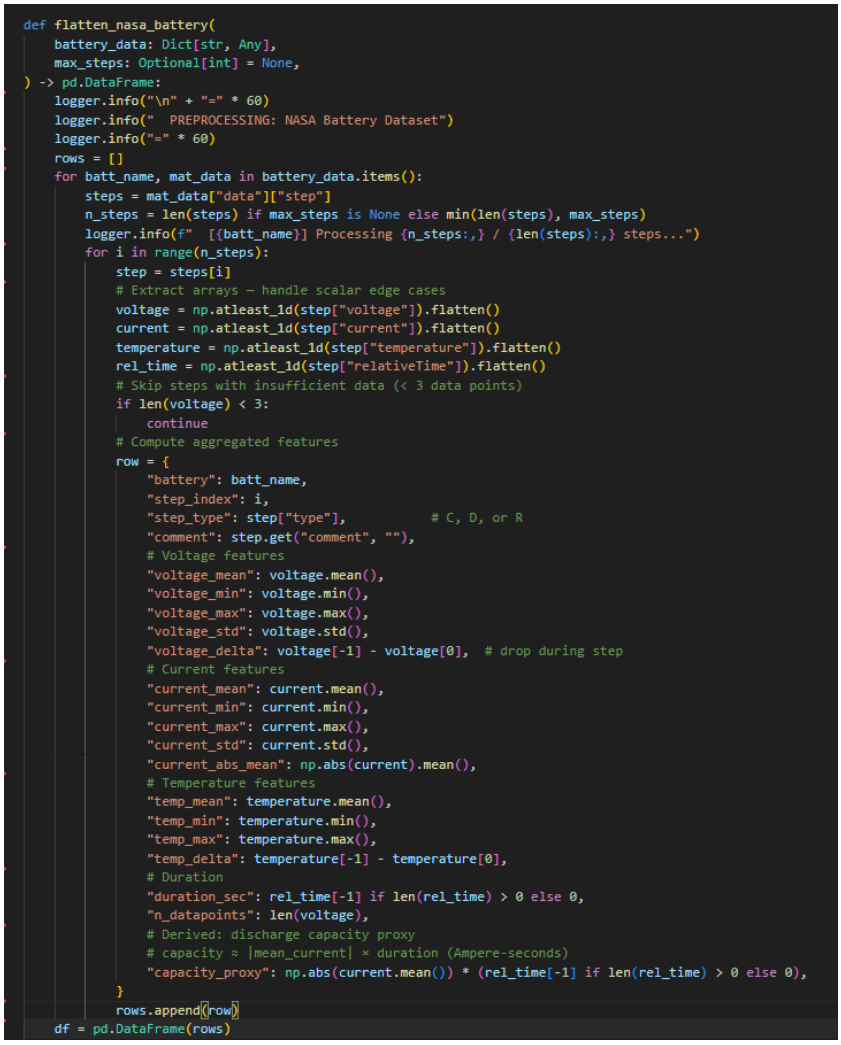
Vehicle telemetry, driver behaviour, battery, and route datasets were loaded and transformed into a unified format before being used by the Energy Management System (EMS). The preprocessing pipeline removed duplicate records, handled missing values, detected and clipped statistical outliers, encoded categorical variables, standardized numerical features, and generated additional engineered features required for machine learning. Specialized preprocessing was also performed for IMU sensor measurements and NASA battery datasets to derive acceleration, gyroscope, battery health, and degradation features used by the predictive EMS controller.

```python
def preprocess_behavior_dataset(df: pd.DataFrame) -> Tuple[pd.DataFrame, LabelEncoder, StandardScaler]:
    logger.info("\n" + "=" * 60)
    logger.info("  PREPROCESSING: Driver Behavior Dataset")
    logger.info("=" * 60)
    name = "Behavior"
    # 1. Duplicates
    df = remove_duplicates(df, name)
    # 2. Missing values
    df = handle_missing_values(df, name=name)
    # 3. Outlier detection
    numeric_cols = ["speed_kmph", "accel_x", "accel_y", "brake_pressure",
                    "steering_angle", "throttle", "lane_deviation",
                    "headway_distance", "reaction_time"]
    detect_outliers_iqr(df, columns=numeric_cols, name=name)
    # 4. Clip extreme outliers (only truly extreme ones, 3×IQR)
    df = clip_outliers(df, columns=numeric_cols, factor=3.0, name=name)
    # 5. Generate synthetic efficiency target
    df = generate_synthetic_efficiency(df)
    # 6. Encode behavior labels
    le = LabelEncoder()
    df["behavior_encoded"] = le.fit_transform(df["behavior_label"])
    logger.info(f"  [{name}] Label encoding: {dict(zip(le.classes_, le.transform(le.classes_)))}")
    # 7. Scale numeric features
    feature_cols = numeric_cols.copy()
    scaler = StandardScaler()
    df_scaled = df.copy()
    df_scaled[feature_cols] = scaler.fit_transform(df[feature_cols])
    # Keep both scaled and unscaled versions
    # The unscaled version is useful for interpretability
    for col in feature_cols:
        df[f"{col}_scaled"] = df_scaled[col]
    logger.info(f"  [{name}] Scaled {len(feature_cols)} numeric features (StandardScaler)")
    logger.info(f"  [{name}] Final shape: {df.shape}")
    return df, le, scaler
```

Figure A.3.1: Driver Behaviour Dataset Preprocessing Pipeline

```python
def engineer_imu_features(df: pd.DataFrame) -> pd.DataFrame:
    # Acceleration magnitude (total g-force)
    df["acc_magnitude"] = np.sqrt(df["AccX"]**2 + df["AccY"]**2 + df["AccZ"]**2)
    # Gyroscope magnitude (total angular velocity)
    df["gyro_magnitude"] = np.sqrt(df["GyroX"]**2 + df["GyroY"]**2 + df["GyroZ"]**2)
    # Jerk (rate of acceleration change) — computed as difference
    # High jerk = sudden driving inputs = aggressive behavior
    df["jerk_x"] = df["AccX"].diff().fillna(0)
    df["jerk_y"] = df["AccY"].diff().fillna(0)
    df["jerk_z"] = df["AccZ"].diff().fillna(0)
    df["jerk_magnitude"] = np.sqrt(df["jerk_x"]**2 + df["jerk_y"]**2 + df["jerk_z"]**2)
    logger.info("  [Telemetry] Engineered features: acc_magnitude, gyro_magnitude, jerk_x/y/z, jerk_magnitude")
    return df
```

Figure A.3.2: IMU Telemetry Feature Engineering



```python
def flatten_nasa_battery(
    battery_data: Dict[str, Any],
    max_steps: Optional[int] = None,
) -> pd.DataFrame:
    logger.info("\n" + "=" * 60)
    logger.info("  PREPROCESSING: NASA Battery Dataset")
    logger.info("=" * 60)
    rows = []
    for batt_name, mat_data in battery_data.items():
        steps = mat_data["data"]["step"]
        n_steps = len(steps) if max_steps is None else min(len(steps), max_steps)
        logger.info(f"  [{batt_name}] Processing {n_steps:,} / {len(steps):,} steps...")
        for i in range(n_steps):
            step = steps[i]
            # Extract arrays — handle scalar edge cases
            voltage = np.atleast_1d(step["voltage"]).flatten()
            current = np.atleast_1d(step["current"]).flatten()
            temperature = np.atleast_1d(step["temperature"]).flatten()
            rel_time = np.atleast_1d(step["relativeTime"]).flatten()
            # Skip steps with insufficient data (< 3 data points)
            if len(voltage) < 3:
                continue
            # Compute aggregated features
            row = {
                "battery": batt_name,
                "step_index": i,
                "step_type": step["type"],           # C, D, or R
                "comment": step.get("comment", ""),
                # Voltage features
                "voltage_mean": voltage.mean(),
                "voltage_min": voltage.min(),
                "voltage_max": voltage.max(),
                "voltage_std": voltage.std(),
                "voltage_delta": voltage[-1] - voltage[0],  # drop during step
                # Current features
                "current_mean": current.mean(),
                "current_min": current.min(),
                "current_max": current.max(),
                "current_std": current.std(),
                "current_abs_mean": np.abs(current).mean(),
                # Temperature features
                "temp_mean": temperature.mean(),
                "temp_min": temperature.min(),
                "temp_max": temperature.max(),
                "temp_delta": temperature[-1] - temperature[0],
                # Duration
                "duration_sec": rel_time[-1] if len(rel_time) > 0 else 0,
                "n_datapoints": len(voltage),
                # Derived: discharge capacity proxy
                # capacity ≈ |mean_current| × duration (Ampere-seconds)
                "capacity_proxy": np.abs(current.mean()) * (rel_time[-1] if len(rel_time) > 0 else 0),
            }
            rows.append(row)
    df = pd.DataFrame(rows)
```

Figure A.3.3: NASA Battery Dataset Processing and Feature Extraction

```python
def engineer_features(df: pd.DataFrame) -> pd.DataFrame:
    logger.info("Engineering EMS features...")
    # 1. Flags
    df["high_power_flag"] = (df["power_required_kw"] > HIGH_POWER_KW).astype(int)
    df["low_soc_flag"] = (df["battery_soc"] < SOC_LOW).astype(int)
    df["critical_soc_flag"] = (df["battery_soc"] < SOC_CRITICAL).astype(int)
    df["is_urban"] = (df["current_segment"] == 'urban').astype(int)
    df["is_highway"] = (df["current_segment"] == 'highway').astype(int)
    df["is_stop_go"] = (df["current_segment"] == 'stop_go').astype(int)
    df["regen_candidate"] = ((df["braking"] == True) &
                             (df["regen_available"] == True) &
                             (df["speed"] > REGEN_MIN_SPEED)).astype(int)
    df["next_is_highway"] = (df["next_segment"] == 'highway').astype(int)
    df["next_is_traffic"] = df["next_segment"].isin(['traffic', 'stop_go']).astype(int)
    # 2. Continuous features
    df["load_intensity_score"] = (df["power_required_kw"] + df["aux_load_kw"]) / 120.0
    # Thermal and grade
    df["thermal_stress_flag"] = ((df["battery_temp"] > BATTERY_TEMP_MAX) |
                                 (df["battery_temp"] < BATTERY_TEMP_MIN)).astype(int)
    df["uphill_flag"] = (df["grade_angle"] > 3.0).astype(int)
    df["downhill_flag"] = (df["grade_angle"] < -3.0).astype(int)
    # Ratios and interactions
    df["power_per_speed"] = df["power_required_kw"] / (df["speed"] + 1e-3)
    df["soc_x_power"] = df["battery_soc"] * df["power_required_kw"]
    logger.info(f"Engineered features added. New shape: {df.shape}")
    return df
```

Figure A.3.4: EMS Feature Engineering and State Variable Generation

A.3.2 Demand Forecasting Module

A machine learning-based demand forecasting module was developed to predict future vehicle power demand, speed, and acceleration using recent telemetry history. The module applies rolling-window temporal feature engineering together with Random Forest regression models to estimate future driving conditions. These predictions provide advisory information to the supervisory Energy Management System (EMS), enabling proactive operating mode selection and improved energy management performance.

This code initializes the `DemandForecaster` class and loads the trained Random Forest regression models for power demand, vehicle speed, and acceleration prediction. The trained models are stored locally and loaded dynamically during runtime.

```python
class DemandForecaster:
    """
    Predicts future speed, acceleration, and power demand using rolling window history.
    """
    def __init__(self):
        self.power_model_path = MODELS_DIR / "power_forecaster.pkl"
        self.speed_model_path = MODELS_DIR / "speed_forecaster.pkl"
        self.accel_model_path = MODELS_DIR / "accel_forecaster.pkl"

        self.power_model = None
        self.speed_model = None
        self.accel_model = None
        self.load_models()

    def load_models(self) -> bool:
        """Loads trained regressor models from models/ directory."""
        try:
            if self.power_model_path.exists():
                self.power_model = joblib.load(self.power_model_path)
            if self.speed_model_path.exists():
                self.speed_model = joblib.load(self.speed_model_path)
            if self.accel_model_path.exists():
                self.accel_model = joblib.load(self.accel_model_path)

            logger.info("Successfully loaded all demand forecasting models.")
            return True
        except Exception as e:
            logger.error(f"Error loading demand forecasting models: {e}")
            return False
```

Figure A.3.5: Demand Forecasting Module Initialization and Model Loading

This section implements the core forecasting algorithm. Historical telemetry is transformed into lag-based temporal features, categorical variables are encoded, feature vectors are prepared, and the trained regression models predict future vehicle power demand, speed, and acceleration for the supervisory EMS.

```python
def predict_future(self, df_history: pd.DataFrame) -> Tuple[float, float, float]:
    if self.power_model is None or self.speed_model is None or self.accel_model is None:
        # Try to load models dynamically if not loaded
        if not self.load_models():
            # Fallback to current values if models are missing
            last_row = df_history.iloc[-1]
            logger.warning("DemandForecaster models not trained. Using current state fallback.")
            return (float(last_row["power_required_kw"]),
                    float(last_row["speed"]),
                    float(last_row["acceleration"]))
    # Prepare the single row (last row of lagged df)
    # First build lags on the history df
    lagged_df = create_lags(df_history, ["speed", "acceleration", "power_required_kw"], n_lags=5)
    last_row_df = lagged_df.iloc[[-1]].copy()
    # Encode categorical columns
    if "traffic_condition" in last_row_df.columns:
        last_row_df["traffic_condition"] = last_row_df["traffic_condition"].map(TRAFFIC_MAP).fillna(0).astype(int)
    if "current_segment" in last_row_df.columns:
        last_row_df["current_segment"] = last_row_df["current_segment"].map(SEGMENT_MAP).fillna(0).astype(int)
    if "next_segment" in last_row_df.columns:
        last_row_df["next_segment"] = last_row_df["next_segment"].map(SEGMENT_MAP).fillna(0).astype(int)
    # Ensure boolean columns are numeric
    for col in ["regen_available", "braking"]:
        if col in last_row_df.columns:
            last_row_df[col] = last_row_df[col].astype(int)
    # Features used for training (must match exactly)
    feature_cols = [
        "speed", "acceleration", "power_required_kw", "battery_soc", "battery_temp",
        "aux_load_kw", "grade_angle", "regen_available", "braking",
        "traffic_condition", "current_segment", "next_segment",
        "speed_lag_1", "speed_lag_2", "speed_lag_3", "speed_lag_4", "speed_lag_5",
        "acceleration_lag_1", "acceleration_lag_2", "acceleration_lag_3", "acceleration_lag_4", "acceleration_lag_5",
        "power_required_kw_lag_1", "power_required_kw_lag_2", "power_required_kw_lag_3", "power_required_kw_lag_4", "power_required_kw_lag_5"
    ]
    # Ensure all columns exist, if not fill with 0.0
    for col in feature_cols:
        if col not in last_row_df.columns:
            last_row_df[col] = 0.0
    # Align features
    X = last_row_df[feature_cols]
    pred_power = float(self.power_model.predict(X)[0])
    pred_speed = float(self.speed_model.predict(X)[0])
    pred_accel = float(self.accel_model.predict(X)[0])
    # INTEGRATION POINT — Alex: replace with real power_required_kw or transient load models
    return pred_power, pred_speed, pred_accel
```

Figure A.3.6: Future Vehicle Demand Prediction Algorithm

This code trains three Random Forest regression models using trip-level data partitioning to prevent temporal data leakage. The implementation generates lag features, constructs future prediction targets, evaluates model performance using MAE, RMSE, and R^2 metrics, and stores the trained models for deployment within the predictive EMS.

```python
def train_demand_forecaster(data_path: Path, model_dir: Path, n_lags: int = 5, forecast_step: int = 5) -> Dict[str, Any]:
    logger.info(f"Loading data from {data_path} to train demand forecaster...")
    df = pd.read_csv(data_path)
    # 1. Generate lagged features within trip groups to prevent leakage
    df_features = create_lags(df, ["speed", "acceleration", "power_required_kw"], n_lags=n_lags)
    # 2. Map categorical columns
    df_features["traffic_condition"] = df_features["traffic_condition"].map(TRAFFIC_MAP).fillna(0).astype(int)
    df_features["current_segment"] = df_features["current_segment"].map(SEGMENT_MAP).fillna(0).astype(int)
    df_features["next_segment"] = df_features["next_segment"].map(SEGMENT_MAP).fillna(0).astype(int)
    # Ensure boolean columns are numeric
    for col in ["regen_available", "braking"]:
        if col in df_features.columns:
            df_features[col] = df_features[col].astype(int)
    # 3. Define feature columns
    feature_cols = [
        "speed", "acceleration", "power_required_kw", "battery_soc", "battery_temp",
        "aux_load_kw", "grade_angle", "regen_available", "braking",
        "traffic_condition", "current_segment", "next_segment",
        "speed_lag_1", "speed_lag_2", "speed_lag_3", "speed_lag_4", "speed_lag_5",
        "acceleration_lag_1", "acceleration_lag_2", "acceleration_lag_3", "acceleration_lag_4", "acceleration_lag_5",
        "power_required_kw_lag_1", "power_required_kw_lag_2", "power_required_kw_lag_3", "power_required_kw_lag_4", "power_required_kw_lag_5"
    ]
    # 4. Generate future targets within each trip_id group to prevent lookahead leak across trips
    grouped = df_features.groupby("trip_id")
    df_features["target_power"] = grouped["power_required_kw"].shift(-forecast_step)
    df_features["target_speed"] = grouped["speed"].shift(-forecast_step)
    df_features["target_accel"] = grouped["acceleration"].shift(-forecast_step)
    # Drop rows where target is NaN (which occur at the end of each trip due to shift)
    df_clean = df_features.dropna(subset=["target_power", "target_speed", "target_accel"]).copy()
    X = df_clean[feature_cols]
    y_power = df_clean["target_power"]
    y_speed = df_clean["target_speed"]
    y_accel = df_clean["target_accel"]
    # Perform a trip-level split or randomized row split (trip-level split is more robust for sequences)
    # Since each trip has a distinct trip_id, we can split trips:
    unique_trips = df_clean["trip_id"].unique()
    np.random.seed(RANDOM_STATE)
    np.random.shuffle(unique_trips)
    split_idx = int(len(unique_trips) * (1.0 - TEST_SIZE))
    train_trips = unique_trips[:split_idx]
    test_trips = unique_trips[split_idx:]
    train_mask = df_clean["trip_id"].isin(train_trips)
    test_mask = df_clean["trip_id"].isin(test_trips)
    X_train, X_test = X[train_mask], X[test_mask]
    y_power_train, y_power_test = y_power[train_mask], y_power[test_mask]
    y_speed_train, y_speed_test = y_speed[train_mask], y_speed[test_mask]
    y_accel_train, y_accel_test = y_accel[train_mask], y_accel[test_mask]
    logger.info(f"Split data into {len(train_trips)} train trips and {len(test_trips)} test trips.")
    logger.info(f"Train shape: {X_train.shape}, Test shape: {X_test.shape}")
    # 5. Train Random Forests (lightweight for fast training & stable inference)
    # Using small max_depth to guarantee generalization and avoid overfitting
    model_params = {"n_estimators": 50, "max_depth": 8, "random_state": RANDOM_STATE, "n_jobs": -1}
    logger.info("Training power demand forecaster...")
    power_rf = RandomForestRegressor(**model_params)
    power_rf.fit(X_train, y_power_train)
    pred_power = power_rf.predict(X_test)
    mae_power = mean_absolute_error(y_power_test, pred_power)
    rmse_power = np.sqrt(mean_squared_error(y_power_test, pred_power))
    r2_power = r2_score(y_power_test, pred_power)
    logger.info(f"  Power Forecaster: MAE={mae_power:.3f} kW, RMSE={rmse_power:.3f} kW, R2={r2_power:.3f}")
    logger.info("Training speed forecaster...")
    speed_rf = RandomForestRegressor(**model_params)
    speed_rf.fit(X_train, y_speed_train)
    pred_speed = speed_rf.predict(X_test)
    mae_speed = mean_absolute_error(y_speed_test, pred_speed)
    rmse_speed = np.sqrt(mean_squared_error(y_speed_test, pred_speed))
    r2_speed = r2_score(y_speed_test, pred_speed)
    logger.info(f"  Speed Forecaster: MAE={mae_speed:.3f} km/h, RMSE={rmse_speed:.3f} km/h, R2={r2_speed:.3f}")
    logger.info("Training acceleration forecaster...")
    accel_rf = RandomForestRegressor(**model_params)
    accel_rf.fit(X_train, y_accel_train)
    pred_accel = accel_rf.predict(X_test)
    mae_accel = mean_absolute_error(y_accel_test, pred_accel)
    rmse_accel = np.sqrt(mean_squared_error(y_accel_test, pred_accel))
    r2_accel = r2_score(y_accel_test, pred_accel)
    logger.info(f"  Acceleration Forecaster: MAE={mae_accel:.3f} m/s², RMSE={rmse_accel:.3f} m/s², R2={r2_accel:.3f}")
```

Figure A.3.7: Random Forest Demand Forecaster Training Pipeline

A.3.3 Hybrid Decision Engine

The Hybrid Decision Engine integrates machine learning predictions with deterministic supervisory rules and safety validation to select an appropriate vehicle operating mode.

Initializes the Hybrid Decision Engine by loading the trained ML models, rule engine, safety layer, and telemetry history buffer.

```python
class HybridDecisionEngine:
    def __init__(self, model_name: str = "ems_best_model.pkl"):
        self.rule_engine = RuleBasedEMS()
        self.predictive_controller = PredictiveEMSController()
        self.safety_layer = SafetyOverrideLayer()
        self.demand_forecaster = DemandForecaster()
        # Load ML Assets
        model_path = MODELS_DIR / model_name
        encoder_path = MODELS_DIR / "ems_label_encoder.pkl"
        if not model_path.exists() or not encoder_path.exists():
            raise FileNotFoundError(
                f"Missing ML models in {MODELS_DIR}. Run run_ems.py --train first."
            )
        self.model = joblib.load(model_path)
        self.label_encoder = joblib.load(encoder_path)
        # Load expected feature names (saved during training)
        feature_path = MODELS_DIR / "ems_feature_names.pkl"
        if feature_path.exists():
            self.expected_features = joblib.load(feature_path)
        else:
            self.expected_features = None
        # State history for online temporal feature engineering & demand forecasting
        self.state_history = deque(maxlen=15)
    def _update_history(self, state: VehicleState):
        """Append state to history, padding with duplicates if history is empty."""
        if len(self.state_history) == 0:
            for _ in range(15):
                self.state_history.append(state)
        else:
            self.state_history.append(state)
    def _get_history_dataframe(self) -> pd.DataFrame:
        """Convert deque of VehicleState to DataFrame."""
        rows = []
        for s in self.state_history:
            row_dict = {k: v for k, v in s.__dict__.items() if k != "timestamp"}
            rows.append(row_dict)
        return pd.DataFrame(rows)
    def _prepare_features(self, df_history: pd.DataFrame,
                          advisory: Optional[PredictiveAdvisory] = None) -> pd.DataFrame:
        """Apply the exact feature engineering used in training."""
```

Figure A.3.8: Hybrid Decision Engine Initialization and Model Loading

Implements the ML-based EMS workflow, including feature preparation, operating mode prediction, and safety validation.

```python
def decide(self, state: VehicleState,
           lookahead: list = None,
           distance_remaining: float = 10.0,
           grade_ahead: float = 0.0) -> EMSDecision:
    # 1. Update history
    self._update_history(state)
    # 2. Predictive Advisory
    if lookahead is None:
        lookahead = [state.next_segment]
    advisory = self.predictive_controller.advise(
        state, lookahead, distance_remaining, grade_ahead
    )
    # 3. Prepare ML Input from history
    df_history = self._get_history_dataframe()
    # Suppress logging for row-by-row simulation
    prep_logger = logging.getLogger("src.preprocessing")
    prev_level = prep_logger.getEffectiveLevel()
    prep_logger.setLevel(logging.WARNING)
    X = self._prepare_features(df_history, advisory)
    prep_logger.setLevel(prev_level)
    # 4. ML Prediction
    pred_idx = self.model.predict(X)[0]
    ml_mode = self.label_encoder.inverse_transform([pred_idx])[0]
    if hasattr(self.model, "predict_proba"):
        probs = self.model.predict_proba(X)[0]
        confidence = float(probs[pred_idx])
    else:
        confidence = 1.0
    # 5. Safety Override Check using SafetyOverrideLayer
    is_safe, violation = self.safety_layer.check_safety(ml_mode, state)
    if is_safe:
        # Build advisory reason string
        adv_notes = []
        if advisory.preserve_battery:
            adv_notes.append("preserve_battery")
        if advisory.prepare_regen:
            adv_notes.append("prepare_regen")
        if advisory.prefer_engine:
            adv_notes.append("prefer_engine")
        if advisory.charge_sustain:
            adv_notes.append("charge_sustain")
        adv_str = f" [Advisory: {', '.join(adv_notes)}]" if adv_notes else ""
        return EMSDecision(
            mode=ml_mode,
            confidence=confidence,
            rule_triggered="ML_PRIMARY",
            reason=(f"ML selected {ml_mode} ({confidence*100:.1f}% conf)."
                    f"{adv_str}"),
            source="ML",
            input_state=state
        )
    else:
        # 6. Fallback to Deterministic Rules
        logger.debug(f"SAFETY OVERRIDE: {violation}")
        rule_decision = self.rule_engine.decide(state)
        rule_decision.source = "HYBRID_OVERRIDE"
        rule_decision.reason = (f"OVERRIDE: {violation} | "
                                f"Fallback -> {rule_decision.reason}")
        return rule_decision
```

Figure A.3.9: Machine Learning-Based EMS Decision-Making Workflow

Implements the rule-based EMS controller that selects the operating mode using vehicle, battery, traffic, and route conditions.

```python
def select_mode(
    telemetry: dict,
    traffic_state: TrafficState,
    soc_plan: SOCPlan,
    battery_health: BatteryHealthState,
    predicted_demand_kw: float,
    regen_result: RegenResult,
) -> EMSMode:
    speed = telemetry.get("speed_kmh", 0.0)
    soc = telemetry.get("soc", 0.0)
    soc_frac = soc / 100.0 if soc > 1.0 else soc
    power_demand = telemetry.get("power_demand_kw", 0.0)
    throttle = telemetry.get("throttle", 0.0)
    current_segment = telemetry.get("current_segment", "urban")
    next_segment = telemetry.get("next_segment", "urban")
    position = telemetry.get("position_km", 0.0)
    rpm = telemetry.get("rpm", 0.0)
    # 1. Regenerative Braking
    if power_demand < 0.0:
        if soc_frac >= MAX_SOC:
            logger.info("RULE6_REGEN: Regen disabled - battery full (SOC=%.2f)", soc_frac)
        else:
            clamped_regen = min(abs(power_demand), MOTOR_PEAK_POWER)
            # 1 Wh = 1/3600 kWh
            recovered_wh = clamped_regen * (1.0 / 3600.0) * 1000.0
            logger.info("RULE6_REGEN: Regenerative braking active. Recovered Wh: %.4f", recovered_wh)
            return EMSMode.REGEN
    # 2. Maximum Acceleration
    if throttle >= 0.90:
        logger.info("RULE7_MAX_ACCEL: Maximum acceleration requested.")
        return EMSMode.HYBRID_ASSIST
    # 3. Low SOC Recovery
    if soc_frac <= 0.25:
        logger.info("RULE8_SOC_RECOVERY: Low SOC (SOC=%.2f) recovery active.", soc_frac)
        return EMSMode.CHARGE_SUSTAIN
    # 4. Urban Stop-and-Go EV Priority
    if speed < 50.0 and (traffic_state == TrafficState.CONGESTED or traffic_state.value == "CONGESTED") and current_segment.lower() == "urban":
        if soc_frac <= (MIN_SOC + 0.05):
            logger.info("RULE1_SUPPRESSED_LOW_SOC: Urban EV Priority suppressed due to low SOC (SOC=%.2f)", soc_frac)
        else:
            logger.info("RULE1_EV_PRIORITY: Urban stop-and-go prioritising EV operation.")
            return EMSMode.EV
    # 5. SOC Preservation Before Urban Zones
    upcoming_urban = next_segment.lower() in ("urban", "stop_go")
    if upcoming_urban and soc_frac < 0.80 and current_segment.lower() in ("highway", "mixed", "arterial"):
        logger.info("RULE2_SOC_PRESERVE: Upcoming urban zone - preserving SOC (SOC=%.2f).", soc_frac)
        return EMSMode.CHARGE_SUSTAIN
    # 6. Traffic-Aware Mode Planning
    if traffic_state == TrafficState.CONGESTED or traffic_state.value == "CONGESTED":
        logger.info("RULE3_TRAFFIC_AWARE: High congestion forecast - biasing EV/Hybrid mode.")
        return EMSMode.HYBRID_ASSIST
    # 7. GPS Route Preview Mode Planning
    if current_segment.lower() == "urban":
        logger.info("RULE4_ROUTE_PREVIEW: Urban segment - favouring EV mode.")
        return EMSMode.EV
    elif current_segment.lower() == "highway":
        logger.info("RULE4_ROUTE_PREVIEW: Highway segment - favouring ICE efficiency mode.")
        return EMSMode.ICE
    # 8. ICE Efficiency Sweet Spot (C.RULE5)
    if rpm > 0.0 and (rpm < 2000.0 or rpm > 3000.0):
        logger.info("RULE5_RPM_SWEET_SPOT: Engine outside peak band (%d RPM). Adjusting split.", int(rpm))
        if rpm > 3000.0:
            # Shed load
            return EMSMode.HYBRID_ASSIST
        else:
            # Deactivate or increase load
            return EMSMode.EV
    # Default fallback
    if soc_frac > MIN_SOC:
        return EMSMode.EV
    else:
        return EMSMode.CHARGE_SUSTAIN
```

Figure A.3.10: Supervisory Rule-Based Energy Management Decision Logic

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